

The Physics & Mathematics of the Hydrogen Orbital Visualiser

Project Documentation

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Abstract

This document explains, from the ground up, the mathematics the simulation computes and why each piece exists. It is written for curious readers who want to understand what they are looking at when looking at the quantum probability cloud visualization of the hydrogen atom.

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1 The question the hydrogen atom answers

A hydrogen atom is one proton and one electron. The old “planetary” picture — the electron orbiting like a tiny planet — is wrong. A charged particle moving in a circle radiates energy; a classical electron would spiral into the nucleus in about 10^{-11} seconds. Atoms are stable, so the classical picture must fail.

Quantum mechanics replaces the orbit with a *wavefunction* $\psi(r, \theta, \phi)$: a complex number attached to every point in space. It is not directly observable. What *is* observable is its squared magnitude:

$$|\psi(r, \theta, \phi)|^2 = \text{probability density of finding the electron there.}$$

Key idea

The cloud you see in the visualiser *is* $|\psi|^2$. Each pearl is a “the electron could be here” vote, weighted by probability.

2 The Schrödinger equation and separation of variables

The time-independent Schrödinger equation for the electron in the proton’s Coulomb field is

$$-\frac{\hbar^2}{2m_e} \nabla^2 \psi - \frac{e^2}{4\pi\epsilon_0 r} \psi = E \psi.$$

Because the potential depends only on r (it is spherically symmetric), the equation *separates*. We guess a product solution

$$\psi(r, \theta, \phi) = R(r) \Theta(\theta) \Phi(\phi),$$

substitute, and divide through. The equation splits into three independent ordinary differential equations — one per coordinate. This is the mathematical reason the atom has exactly *three* quantum numbers: one constant of separation emerges from each equation.

$R(r) \rightarrow$ radial equation $\rightarrow$ gives n, ℓ ; Laguerre polynomials,

$\Theta(\theta) \rightarrow$ polar equation $\rightarrow$ gives ℓ, m ; Legendre functions,

$\Phi(\phi) \rightarrow$ azimuthal equation $\rightarrow$ gives m ; $e^{im\phi}$.

The angular pieces $\Theta \cdot \Phi$ combine into the *spherical harmonics* $Y_{\ell m}(\theta, \phi)$.

↓ Rabbit hole: why does separability need spherical symmetry?

Separation of variables only works when the equation “doesn’t mix” the coordinates. The Coulomb potential $-1/r$ depends on r alone — it says nothing about direction. That lets us split “how far” (radial) from “which way” (angular) cleanly. If the potential depended on θ or ϕ (say, in a strong external field), the coordinates would couple and this trick would fail. Spherical symmetry is the gift that makes hydrogen exactly solvable.

3 The three quantum numbers

n — **principal**

$n = 1, 2, 3, \dots$ Sets the energy, $E_n = -13.6 \text{ eV}/n^2$. Larger n means a larger, higher-energy orbital, electron farther from the nucleus.

ℓ — **azimuthal**

$\ell = 0, 1, \dots, n-1$. Sets the *shape* and the total angular momentum $|L| = \hbar\sqrt{\ell(\ell+1)}$.
 $\ell = 0$ (s, sphere), $\ell = 1$ (p, dumbbell), $\ell = 2$ (d, cloverleaf), $\ell = 3$ (f).

m — **magnetic**

$m = -\ell, \dots, 0, \dots, +\ell$. Sets the *orientation* and the z -component of angular momentum $L_z = m\hbar$. Called “magnetic” because it determines the energy splitting in a magnetic field (the Zeeman effect) and, as we shall see, because a complex orbital with $m \neq 0$ carries a real circulating current.

The constraints $\ell < n$ and $|m| \leq \ell$ are not arbitrary: they fall out of requiring the solutions to be finite and single-valued.

4 The radial part: why Laguerre polynomials appear

The radial solutions (in atomic units, $a_0 = 1$) are

$$R_{n\ell}(r) = N_{n\ell} e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho), \quad \rho = \frac{2Zr}{n}.$$

Three factors, each with a job:

- $e^{-\rho/2}$ — exponential decay. Guarantees the electron is *bound*: the probability must vanish at infinity. Without it, ψ would not be normalisable.
- ρ^ℓ — behaviour near the nucleus. For $\ell > 0$ the electron is pushed away from $r = 0$ by the “centrifugal barrier” of angular momentum, so density starts at zero and rises.
- $L_{n-\ell-1}^{2\ell+1}(\rho)$ — the *associated Laguerre polynomial*. This part oscillates, creating the alternating shells of high and low density (the *radial nodes*).

The code computes L with the stable three-term recurrence

$$\begin{aligned} L_0^\alpha(x) &= 1, \\ L_1^\alpha(x) &= 1 + \alpha - x, \\ L_k^\alpha(x) &= \frac{(2k-1+\alpha-x)L_{k-1}^\alpha(x) - (k-1+\alpha)L_{k-2}^\alpha(x)}{k}, \quad k \geq 2. \end{aligned}$$

↓ Rabbit hole: what *is* a Laguerre polynomial?

Laguerre polynomials $L_n(x)$ are a family of polynomials that are *orthogonal* on $[0, \infty)$ with respect to the weight e^{-x} :

$$\int_0^\infty e^{-x} L_m(x) L_n(x) dx = 0 \quad \text{whenever } m \neq n.$$

“Orthogonal” is the key word. Just as the unit vectors $\hat{x}, \hat{y}, \hat{z}$ are mutually perpendicular and let you build any 3D vector from independent components, an orthogonal polynomial family lets you build any reasonable function on $[0, \infty)$ from independent “polynomial directions.” They are the natural coordinate system for problems on a half-line with exponential decay — exactly an electron bound near a nucleus.

The *associated* Laguerre polynomials $L_n^\alpha(x)$ generalise this to the weight $x^\alpha e^{-x}$; the superscript $\alpha = 2\ell + 1$ tilts the weighting to account for the angular-momentum barrier. Crucially, they are not *chosen* — they are *forced*. When you substitute $R = e^{-\rho/2} \rho^\ell f(\rho)$ into the radial equation, the leftover equation for f is exactly the associated Laguerre differential equation. Only its polynomial solutions keep ψ finite; anything else blows up at infinity.

Key idea

The polynomial degree is $n - \ell - 1$, and that number equals the count of *radial nodes*. This is why higher n (at fixed ℓ) shows more concentric shells.

5 Normalisation and the Gamma function

The radial normalisation constant is

$$N_{n\ell} = \sqrt{\left(\frac{2Z}{n}\right)^3 \frac{(n - \ell - 1)!}{2n(n + \ell)!}} = \sqrt{\left(\frac{2Z}{n}\right)^3 \frac{\Gamma(n - \ell)}{2n\Gamma(n + \ell + 1)}}.$$

It is chosen so that $\int_0^\infty r^2 |R_{n\ell}(r)|^2 dr = 1$ — i.e. the electron is somewhere with probability 1.

↓ Rabbit hole: why the Gamma function instead of a factorial loop?

The Gamma function satisfies $\Gamma(k+1) = k!$ for non-negative integers, so $(n+\ell)! = \Gamma(n+\ell+1)$ and $(n-\ell-1)! = \Gamma(n-\ell)$. Two reasons to use it:

1. **Range and safety.** Γ returns a floating-point value and handles large arguments smoothly, avoiding integer overflow (14! already overflows a 32-bit integer).
2. **Generality.** Γ *interpolates* the factorial to all real (and complex) arguments,

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt,$$

which is smooth everywhere except the non-positive integers.

⇓ Deeper rabbit hole: why is $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ “the” factorial?

Integrate by parts: $\Gamma(x+1) = \int_0^\infty t^x e^{-t} dt = [-t^x e^{-t}]_0^\infty + x \int_0^\infty t^{x-1} e^{-t} dt = x \Gamma(x)$. So Γ obeys the *same recurrence* as the factorial, $\Gamma(x+1) = x \Gamma(x)$, with $\Gamma(1) = 1$. Any smooth, log-convex function with that recurrence and that starting value is forced to be the factorial’s unique extension (the Bohr–Mollerup theorem). That is what makes Γ not just a generalisation but *the* natural one.

Watch out

A common bug is writing $\Gamma(n+\ell)$ where $(n+\ell)! = \Gamma(n+\ell+1)$ is meant — off by a factor of $(n+\ell)$. This only rescales the absolute density, so it is invisible when colours are normalised relative to a maximum, but it is still wrong.

6 The angular part: why Legendre functions appear

The angular solutions are the spherical harmonics

$$Y_{\ell m}(\theta, \phi) = N_{\ell m} P_\ell^m(\cos \theta) e^{im\phi}, \quad N_{\ell m} = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}}.$$

- $e^{im\phi}$ — the azimuthal (ϕ) dependence. Single-valuedness (going once around, $\phi \rightarrow \phi + 2\pi$, must return the same value) *forces m to be an integer*.
- $P_\ell^m(\cos \theta)$ — the *associated Legendre function*. This polar (θ) dependence creates the *angular nodes* (nodal cones and planes) that give orbitals their lobed shapes.

↓ Rabbit hole: what *is* a Legendre polynomial?

Legendre polynomials $P_\ell(x)$ are the natural orthogonal family on $[-1, 1]$ with weight 1:

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0 \quad \text{whenever } m \neq n.$$

The interval $[-1, 1]$ is exactly the range of $\cos \theta$ as θ runs from 0 (north pole) to π (south pole), so Legendre polynomials are the natural language for functions of *latitude* on a sphere. They arise from Laplace's equation $\nabla^2 f = 0$ in spherical coordinates — the same Laplacian ∇^2 that appears in the Schrödinger kinetic term. This is why they show up all over physics (gravity, electrostatics, heat flow on a sphere, quantum mechanics): all share the Laplacian. The *associated* Legendre functions P_ℓ^m handle $m \neq 0$ via

$$P_\ell^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x).$$

The factor $(1 - x^2)^{m/2} = (\sin \theta)^m$ pushes the function to zero at the poles for $m \neq 0$ — which is why, e.g., a p_x orbital has no density along the axis perpendicular to its lobes.

Key idea

The number of angular nodes is ℓ . Combined with the $n - \ell - 1$ radial nodes, the *total* node count is always $n - 1$ — an exact bookkeeping rule you can verify by eye in the visualiser.

7 Spherical harmonics: real vs. complex

The textbook harmonics $Y_{\ell m}$ are *complex* (they contain $e^{im\phi}$) and have azimuthally symmetric $|Y|^2$ (rings and shells), not the familiar lobes. Chemists therefore use *real* combinations of $+m$ and $-m$:

$$\text{real, } m > 0 : \quad \propto P_\ell^m(\cos \theta) \cos(m\phi),$$

$$\text{real, } m < 0 : \quad \propto P_\ell^{|m|}(\cos \theta) \sin(|m|\phi).$$

These are the $p_x, p_y, d_{xy}, d_{z^2}, \dots$ orbitals. The code implements exactly this mapping (with a $\sqrt{2}$ amplitude factor).

A subtlety that matters for the animation

A real orbital is an equal mix of $+m$ and $-m$, whose circulating currents *cancel exactly to zero*. So the shapes displayed (real orbitals) have no net current, even though the swirl animation depicts one. The animation corresponds to the *complex* orbital, not the real one shown. See Section 9.

8 From $|\psi|^2$ to particles: sampling and the Jacobian

We want points distributed as $|\psi|^2$. The subtlety is the *volume element*. In Cartesian coordinates $dV = dx dy dz$; in spherical,

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi.$$

The extra factors r^2 and $\sin \theta$ are the *Jacobian* of the coordinate change. They are not optional — and they are the single most common source of “my first orbital plot looks wrong” bugs.

↓ Rabbit hole: what *is* a Jacobian?

When you change coordinates — say from Cartesian (x, y, z) to spherical (r, θ, ϕ) — a small box in one system maps to a differently-sized region in the other. The *Jacobian* is the number that tells you how much a tiny volume stretches or shrinks under the change.

A one-dimensional analogy: substituting $u = 2x$ in an integral gives $dx = \frac{1}{2} du$. That $\frac{1}{2}$ is a one-dimensional Jacobian — the “how much does length rescale” factor. In several dimensions the same idea applies to *volume*, and the factor can vary from point to point. Concretely: consider the spherical shell between radius r and $r + dr$. Its *surface area* grows with r (a big sphere has more skin than a small one), so the shell’s volume is $4\pi r^2 dr$ — the r^2 appears because there is simply *more room* far from the origin. Likewise, near the equator ($\theta = \pi/2$) a strip of latitude covers more area than near the poles ($\theta \rightarrow 0, \pi$), where the strips shrink to points — hence the $\sin \theta$.

↓↓ Deeper rabbit hole: where do $r^2 \sin \theta$ come from, precisely?

The Jacobian is the absolute value of the determinant of the matrix of partial derivatives of (x, y, z) with respect to (r, θ, ϕ) :

$$J = \left| \det \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} \right|.$$

With $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, computing the 3×3 determinant gives, after simplification,

$$J = r^2 \sin \theta.$$

The determinant measures the (signed) volume of the parallelepiped spanned by the three coordinate-tangent vectors; its magnitude is exactly the local volume-scaling factor. So $r^2 \sin \theta$ is not a convention — it is the geometry of spherical coordinates made quantitative.

With the Jacobian in hand, the correct one-dimensional marginal distributions are

$$\begin{aligned} P(r) dr &\propto r^2 |R_{nl}(r)|^2 dr, \\ P(\theta) d\theta &\propto \sin \theta |\Theta(\theta)|^2 d\theta, \\ P(\phi) d\phi &\propto |\Phi(\phi)|^2 d\phi. \end{aligned}$$

Why the r^2 matters

Forgetting the r^2 makes every beginner’s first orbital pile up wrongly at the origin, because $|R|^2$ is often largest there while the actual *probability* (density $\times$ available volume) peaks farther out. This code includes it correctly.

Inverse-CDF sampling

Inverse transform sampling works because the cumulative distribution function $F(x) = \int^x P$ is monotonic from 0 to 1. If u is uniform on $[0, 1]$, then $F^{-1}(u)$ is distributed as P . The code:

1. discretises each axis into a fixed number of slices;
2. computes the marginal PDF at each slice (with the Jacobians above);
3. accumulates a running sum into a normalised CDF array;
4. for each particle, rolls $u \in [0, 1]$ and inverts the CDF with a binary search.

This is $O(N \log \text{RES})$ and vastly more efficient than rejection sampling, which discards $> 99\%$ of trials for diffuse clouds.

9 The probability current — what it really is

The quantum probability current density is

$$\mathbf{J} = \frac{\hbar}{m_e} \text{Im}(\psi^* \nabla \psi).$$

It is the quantum analogue of a fluid flow: it satisfies a continuity equation $\partial_t |\psi|^2 + \nabla \cdot \mathbf{J} = 0$, so it literally describes how probability moves.

For a *complex* hydrogen eigenstate $\psi_{n\ell m}$, the radial and polar parts are real and contribute nothing to $\text{Im}(\psi^* \nabla \psi)$; only the $e^{im\phi}$ factor survives, giving a purely azimuthal current circling the z -axis,

$$J_\phi = \frac{\hbar m}{m_e r \sin \theta} |\psi|^2.$$

Why m is called “magnetic”

A complex $m \neq 0$ orbital carries a genuine, steady circulation of probability around the axis. Circulating charge is a current loop, which produces a magnetic moment — the deep reason m is the *magnetic* quantum number.

⇓ Deeper rabbit hole: if there is a current, why doesn’t the cloud move?

A single eigenstate is a *stationary state*: $|\psi_{n\ell m}|^2$ is independent of time by construction (the time dependence is a global phase $e^{-iEt/\hbar}$ that cancels in $|\psi|^2$). The current $\mathbf{J}$ is non-zero but *steady*: probability circulates through an *unchanging* density, like water going round a closed loop that looks perfectly still. A faithful render of a single eigenstate’s density would therefore be completely static. The swirl in the visualiser is a depiction of the current *field*, not of anything a density-observer would see move.

Three honest caveats.

1. The current exists only for *complex* orbitals; the *real* orbitals shown have zero net current (their $\pm m$ currents cancel).
2. In a stationary state the density does not move (previous box).
3. The code’s animation uses a *softened* angular-speed law ($\dot{\phi} \propto m/(r \sin \theta)$) rather than the exact $\dot{\phi} \propto m/(r^2 \sin^2 \theta)$, which is singular near the axis and the origin and would be numerically unstable and visually violent.

Honest framing

The animation is a legitimate, evocative depiction of a real and usually-ignored phenomenon (the m -driven circulation behind orbital magnetism). It is *not* a literal picture of the displayed orbital’s motion. Present it as “a stylised depiction of the probability current,” and note that the density itself is stationary.

10 Nodes: reading the shape of an orbital

Radial nodes	$n - \ell - 1$	(spherical shells of zero density)
Angular nodes	ℓ	(planes or cones of zero density)
Total nodes	$n - 1$	

States worth trying in the visualiser:

- $(1, 0, 0)$ 1s — no nodes, a solid ball.
- $(2, 0, 0)$ 2s — one radial node: a ball inside a shell.
- $(2, 1, 0)$ 2p_z — one angular node: two lobes along z .
- $(3, 2, 0)$ 3d_{z²} — the famous “donut + lobes.”
- $(4, 3, m)$ 4f — rich lobe structure; a good stress test.

Turning on the cross-section is the best way to *see* radial nodes, which are otherwise hidden inside the outer shells.

11 Atomic units in the code

The code works in *atomic units*: $a_0 = \hbar = m_e = 1$, with physical prefactors dropped. This is standard in atomic physics — it strips the equations of clutter so the *structure* is visible. Distances are in Bohr radii; the zoom radius scales with n^2 . Z is the nuclear charge (1 for hydrogen). Because everything is in atomic units and the flow law is softened, the “speed” and “time step” are visualisation parameters, not physical seconds — another reason to frame the animation as stylised.

Further reading

- D. J. Griffiths, *Introduction to Quantum Mechanics* — the standard first course; excellent on hydrogen and spherical harmonics.
- J. J. Sakurai & J. Napolitano, *Modern Quantum Mechanics* — deeper treatment of angular momentum and currents.
- G. B. Arfken & H. J. Weber, *Mathematical Methods for Physicists* — Laguerre/Legendre polynomials, orthogonality, the Gamma function.